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- Makeup work for missed class due to illness

Question 1: Food industry - List 5 foods, where microbes or their by-products are used. Briefly describe the process and the names of the microbes used.

1. Bakers yeast
    a. Bakers yeast protein is derived from Saccharomyces cerevisiae through fermentation. In the presence of oxygen, the bacteria converts sugars into CO2.
    b. Protein extraction from the bacteria includes a centrifugation step to harvest cells, adding a sample buffer, heating the sample, more centrifugation, and then extracting the proteins.
2. Xanthan Gum
    a. Xanthan gum is derived from Xanthomonas campestris and is produced from carbon through fermentation.
3. Catalase derived from Micrococcus lysodeikticus for use in the manufacture of cheese
    a. Through a pure culture purification process, catalse is derived from Mircococcus lysodeikticus. This catalase is then used to destroy the presence of hydrogen peroxide, which is originally used in the manufacture of cheese to retain moisture.
4. Candida lipolytica for fermentation production of citric acid.
    a. Again, this organism is used in a fermentation process (specifically a carbohydrate based submerged fermentation) to produce citric acid, an ingredient which naturally occurs in some foods and which is added to some canned and jarred foods to keep them fresh longer.
5. Bifidobacterium adolescentis used in dairy yogurt
    a. A culture of this bacteria can be directly added to milk during the yogurt fermentation process to serve as a probiotic in yogurts. Many yogurts have this bacteria in them to improve microbial balance in the gut

Question 2: What is bioremediation? List 5 different areas, and briefly describe the processes used and the names of the microbes used for bioremediation.

Bioremediation is a process that uses living organisms to break down and remove pollutants and hazardous materials and substances from the environment into less toxic products.

1. Cleaning up Soil

    a. Pesticides in soil can be cleaned up by aerobci bacteria like Bacillus, Pseudomonas, Sphingomonas, Flavobacterium, Nocardia, Rhodococcus, and Mycobacterium
        i. These bacteria, in the presence of oxygen, use pesticides as a source of energy. Oxygen serves as an electron acceptor in their metabolic processes and the more oxygen you pump into the system, the more effective the degradation is.
2. Wastewater Treatment
    a. Tetrasphaera, Trichococcus, Candidatus Microthrix, Rhodoferax, Rhodobacter, and Hyphomicrobium, are all microbial cleaners in wastewater treatment
        i. Tatraspaherea for example acts as a phosphorus-accumulating organism and reduces the amount of phosphorus in the water at wastewater treatment plants.
3. Groundwater remediation
    a. One microbe used in groundwater remediation is Dehalococcoides mccartyi, which helps dechlorinate tetrachloroethene and trichloroethene through a process called reductive dechlorination. It essentially removes the chlorine atoms from contaminant molecules until only the ethene remains.
4. Oil spill cleanup
    a. A process called bioaugmentation adds oil-degrading bacteria to the existing microbial population in a spill. This bacteria is known as hydrocarbon-degrading bacteria and one species example is Pseudomonas putida.
5. Composting as a form of bioremediation (reducing landfill waste)
    a. Mesophilic bacteria help produce organic acids that aid the beginning of the composting process and break down the readily degradable components. The heat produced in this process also allows compost temperature to subsequently increase.

Question 3: List two examples each for the use of microbes or their by-products in the medical, agricultural, laundry and alternative energy fields.

Medical

- Antibiotics
- Insulin

Agricultural

- Insecticides to help control insect infestations
- Fungicide to prevent and treat fungal infections in crops

Laundry

- Protease breaking down protein stains
- Lipase breaking down oil stains

Alternative Energy

- Bioethanol as a renewable energy source in place of fossil fuels
- Biohydrogen as a renewable energy source and take CO2 out of the atmosphere

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